

comiccraft-AI
comic story
creator using
Gemini Models

Slide 1 – Application Architecture

Activity 1.2: Define the Architecture of the Application

What is Application Architecture?

- Application architecture defines the overall structure of an application.
- It describes how different components communicate with each other.
- It helps organize the application into manageable modules.
- A good architecture improves performance, security, scalability, and maintenance.

Main Components

- **User Interface (Frontend)**
 - **Application/Business Logic (Backend)**
 - **Database**
 - **External Services/APIs**
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Slide 2 – Architecture Diagram

Three-Tier Architecture

1. Presentation Layer

- Provides the user interface.
- Accepts user input and displays results.

- Examples: Web pages, mobile screens, dashboards.



2. Application/Business Logic Layer

- Processes user requests.
- Implements application rules and functions.
- Communicates between the frontend and database.



3. Data Layer

- Stores and manages application data.
- Handles database operations such as Create, Read, Update, and Delete (CRUD).
- Examples: MySQL, PostgreSQL, MongoDB.

Flow:

User → Frontend → Backend → Database → Backend → Frontend → User

Slide 3 – Key Architectural Components

Frontend

- User-friendly interface
- Input forms and navigation
- Displays application results

Backend

- Handles business logic
- Provides APIs
- Performs authentication and authorization
- Validates user requests

Database

- Stores user and application information
- Maintains data consistency
- Provides secure data access

API / External Services

- Connects the application with external systems.
 - Supports services such as payment, email, maps, or notifications.
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Slide 4 – Benefits of the Proposed Architecture

Advantages

- **Scalability:** Components can be expanded as the number of users increases.
- **Maintainability:** Each layer can be developed and updated independently.
- **Security:** Authentication and data access can be controlled at the backend.
- **Performance:** Proper separation of responsibilities improves efficiency.
- **Reliability:** Problems in one component can be isolated and managed.
- **Reusability:** APIs and business logic can be reused across different platforms.

Conclusion

The proposed **three-tier architecture** provides a structured, secure, scalable, and maintainable foundation for developing the application.